

DOVER AFB, DE, USA

WMO: 724088

Lat: 39.133N

Lon: 75.467W

Elev: 28

StdP: 14.68

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13707

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.9	18.0	-0.6	5.3	17.0	3.4	6.6	20.6	28.5	38.2	26.4	35.2	10.6	310	0.497

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.7	92.4	76.1	90.0	75.3	87.5	74.4	79.8	86.7	78.2	85.1	76.9	83.6	10.7	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.2	146.4	82.3	76.4	137.7	81.2	75.0	131.4	80.2	43.3	86.5	41.6	85.5	40.2	83.7	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
25.0	21.4	19.0	DB	9.0	96.5	5.0	3.0	5.4	98.7	2.5	100.5	-0.4	102.2	-4.0	104.4
			WB	7.5	81.6	4.7	1.8	4.1	82.9	1.3	83.9	-1.3	85.0	-4.7	86.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	56.3	34.3	36.9	44.0	54.3	63.7	72.7	77.3	75.7	69.4	58.4	47.5	39.8	(6)
	DBStd	16.90	10.39	9.59	9.29	8.52	7.70	6.33	4.98	4.81	6.25	7.89	8.14	9.01	(7)
	HDD50	1655	501	379	234	45	1	0	0	0	15	141	339		(8)
	HDD65	4433	953	786	652	338	121	12	0	27	236	526	782		(9)
	CDD50	3939	12	13	48	176	425	680	845	796	581	275	66	22	(10)
	CDD65	1243	0	0	2	18	80	242	380	331	158	31	1	0	(11)
	CDH74	10162	0	0	18	157	656	2008	3501	2669	983	165	4	1	(12)
	CDH80	3494	0	0	3	32	197	695	1382	895	261	29	0	0	(13)
(14) Wind	WSAvg	8.9	10.1	10.2	10.8	10.2	8.4	8.0	7.4	7.1	7.8	8.4	9.0	9.4	(14)
(15) Precipitation	PrecAvg	45.2	3.5	3.0	4.1	3.5	4.0	3.9	4.2	4.5	4.4	3.3	3.2	3.6	(15)
	PrecMax	63.0	7.7	7.0	9.3	8.1	7.8	9.9	14.3	16.1	11.9	9.0	6.5	8.4	(16)
	PrecMin	22.1	0.4	0.6	0.2	0.5	0.6	0.7	1.1	1.2	0.3	0.0	0.6	0.8	(17)
	PrecStd	9.0	1.7	1.6	1.9	1.6	1.9	2.1	2.5	2.7	2.6	2.0	1.8	2.0	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	67.2	69.4	75.9	84.4	89.6	93.9	96.4	94.4	91.2	83.2	73.1	69.0	(19)
		MCWB	61.4	60.8	61.3	68.0	74.4	76.7	77.2	77.8	75.1	71.2	65.0	63.8	(20)
	2%	DB	61.8	62.9	69.4	79.0	85.7	90.3	93.0	90.7	86.6	78.7	68.3	63.4	(21)
		MCWB	58.3	55.6	57.1	65.6	71.9	74.7	76.2	75.8	73.9	68.9	61.6	59.7	(22)
	5%	DB	55.4	56.6	64.0	73.6	81.4	87.3	90.3	88.0	82.4	74.4	64.2	58.8	(23)
		MCWB	51.1	50.3	55.6	62.6	69.7	74.0	75.8	74.9	72.1	66.7	58.9	54.9	(24)
	10%	DB	49.7	51.1	59.0	69.1	77.1	84.0	87.6	85.0	79.4	71.1	60.6	53.9	(25)
		MCWB	45.7	45.2	52.0	60.2	67.6	72.8	75.0	74.1	70.5	64.8	55.8	50.6	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	63.4	63.4	64.8	73.3	77.3	80.3	82.8	82.5	78.9	75.2	67.6	65.8	(27)
		MCDB	65.6	68.4	72.2	76.2	84.3	86.1	88.0	87.9	84.7	78.8	70.0	67.0	(28)
	2%	WB	58.7	57.5	60.9	67.9	74.3	77.8	80.2	79.0	76.9	71.8	64.1	60.5	(29)
		MCDB	61.6	60.5	67.0	75.7	81.8	85.6	87.4	86.3	82.2	76.1	66.9	62.8	(30)
	5%	WB	52.3	51.3	57.1	64.9	72.1	76.3	78.5	77.4	75.0	69.2	60.4	55.5	(31)
		MCDB	55.0	55.4	62.1	71.5	79.0	83.8	85.6	84.0	79.8	72.5	63.1	57.8	(32)
	10%	WB	46.0	46.0	53.0	61.9	69.7	74.7	77.2	75.9	73.2	66.1	57.0	50.8	(33)
		MCDB	48.1	49.9	58.8	68.1	75.3	81.5	84.1	81.9	77.7	69.8	59.7	53.0	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	14.8	16.1	17.4	18.6	17.3	16.9	16.7	16.3	16.4	17.2	16.6	15.0	(35)
		MCDBR	22.2	23.8	25.2	24.5	22.1	20.5	19.6	18.8	19.1	19.9	20.8	19.9	(36)
	5% WB	MCWBR	18.7	18.4	16.9	14.1	11.5	9.0	7.4	7.6	9.2	12.1	15.1	17.6	(37)
		MCWBR	21.3	23.0	23.2	21.1	19.3	18.0	16.7	16.1	15.8	16.3	17.7	18.7	(38)
(40) Clear-Sky Solar Irradiance	taub		0.314	0.320	0.368	0.398	0.444	0.499	0.516	0.509	0.421	0.362	0.328	0.312	(40)
		taud	2.466	2.447	2.353	2.298	2.200	2.093	2.060	2.065	2.286	2.456	2.504	2.514	(41)
	Ebn at Noon		272	287	281	279	267	252	246	244	260	268	265	265	(42)
		Edh at Noon	25	30	36	41	46	51	52	50	38	29	24	23	(43)
(44) All-Sky Solar Radiation	RadAvg		653	929	1230	1582	1778	1946	1930	1709	1403	1035	744	573	(44)
	RadStd		56	101	101	121	210	145	88	106	146	106	65	56	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (69 neighbors)	+0.97	N/A	N/A	N/A	+1.10	+1.46	N/A	N/A	+261	+164
		+0.85	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+247	+142

Nomenclature: See separate page